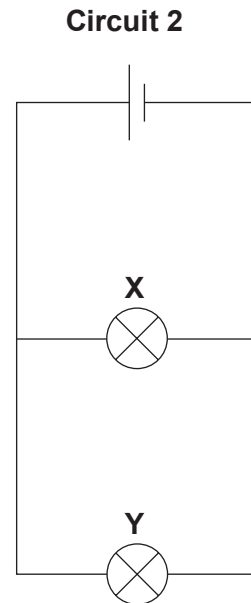
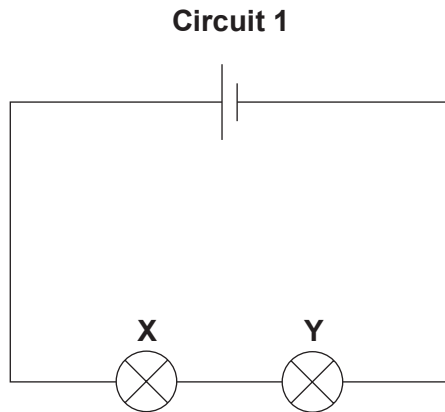


2. A student is studying the properties of different electrical circuits. **X** and **Y** are identical lamps. The student built the following circuits.



- (a) Complete the sentences using the terms in the box.  
The terms can be used once, more than once or not at all.

|              |           |          |             |
|--------------|-----------|----------|-------------|
| series       | less than | parallel | the same as |
| greater than | simple    | current  | voltage     |

- (i) The lamps in **circuit 1** are connected in .....  
The current through lamp **X** is .....the current through lamp **Y**.  
[2]
- (ii) The lamps in **circuit 2** are connected in .....  
The current through lamp **X** is .....the current through lamp **Y**.  
[2]

(b) In **circuit 1** the voltage supplied is 9 V. Each lamp has a resistance of 3 Ω.

(i) Use the equation:

$$\text{current} = \frac{\text{voltage}}{\text{resistance}}$$

to calculate the current in this circuit.

[3]

Current = ..... A

(ii) State what would happen to the current in **circuit 1** if lamp **Y** is removed and the circuit was built with only lamp **X**. [1]

.....

(c) In **circuit 2** the voltage supplied is also 9 V and each bulb has a current of 3 A.

Use the equation

$$\text{power} = \text{voltage} \times \text{current}$$

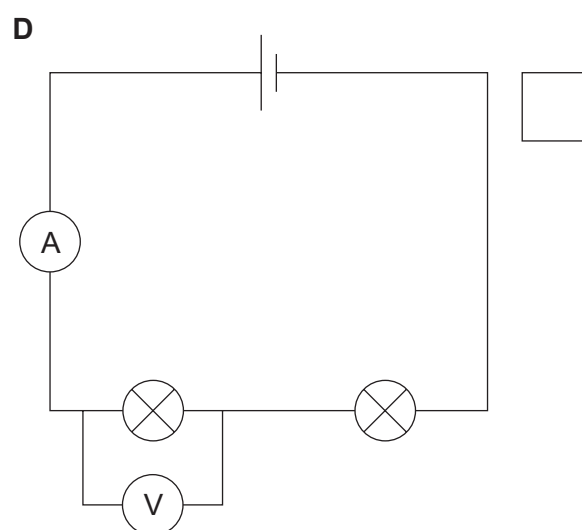
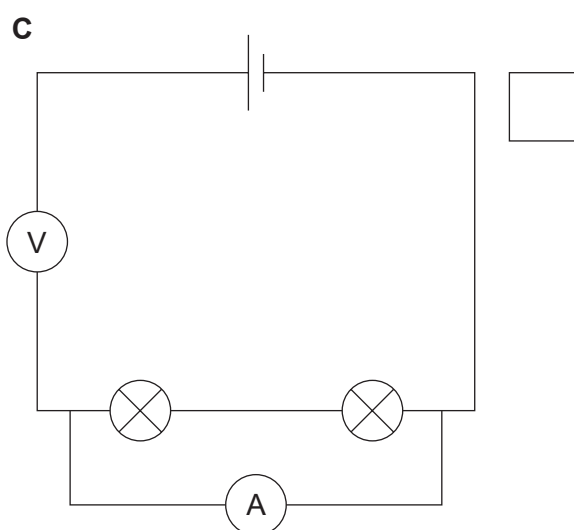
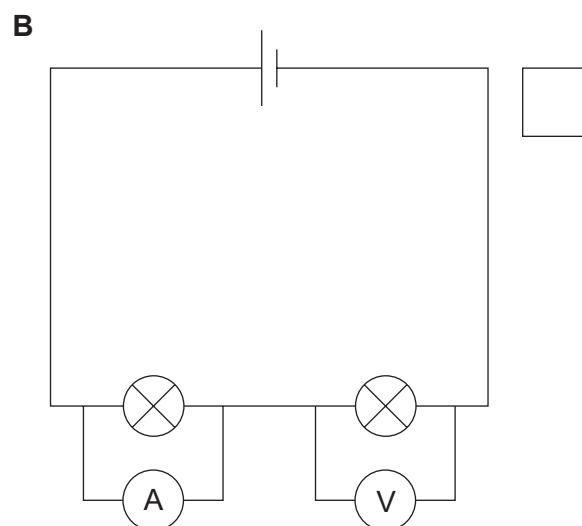
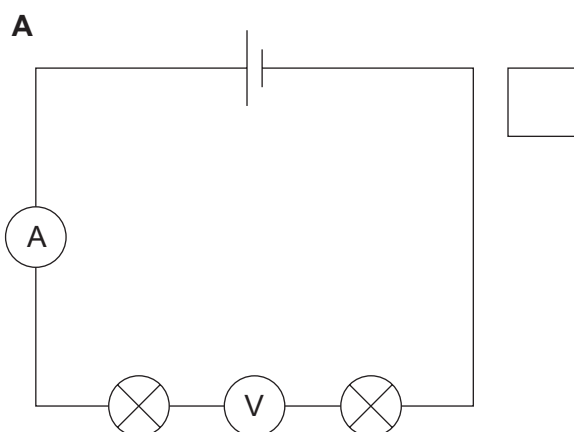
to calculate the power of lamp **X**.

[2]

Power = ..... W

- (d) The student set up a circuit to measure the current and voltage correctly.

Use a tick (✓) to identify the circuit diagram that should have been used to measure the voltage. [1]



- (e) State **two** reasons why lamps are connected in parallel rather than in series in the home. [2]

1. ....
2. ....

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